Experiment 15

Aim

To study the different career pathways available in low-code development and understand how low-code platforms help individuals build software applications, automate business processes, improve productivity, and develop technical skills for employment and entrepreneurship. The study also aims to identify the skills required to pursue a career in low-code development.

Introduction:

Low-code development is an approach to software development that uses visual tools, drag and-drop components, predefined templates, and minimal programming. Platforms such as Zoho Creator, Microsoft Power Apps, and OutSystems enable developers to build applications quickly and automate business processes. Low-code development is becoming useful in organizations that want to reduce development time, improve productivity, and automate repetitive tasks. It creates career opportunities for both technical and nontechnical professionals.

1. **Low-Code Application Developer:**

   A low-code application developer creates business applications using visual development platforms. Responsibilities include designing forms, creating reports, implementing workflows, and testing application functionality. For example, a developer can build a vehicle service management application in Zoho Creator to register vehicles, receive service requests, track repairs, and manage service costs.

2. **Business Process Automation Specialist:**

   These professionals identify repetitive business activities and automate them using workflows, scheduled tasks, notifications, and approval processes. In a vehicle service application, this role could involve automating booking confirmations, service reminders, repair approvals, and customer notifications.

3. **Low-Code Solution Architect**

A solution architect plans the structure of an application, selects suitable technologies, designs integrations, and ensures that the system meets business requirements. For a vehicle service centre, the architect might plan how customer requests, repair details, inventory, payments, and analytics should work together.

4. **Business Analyst:**

Business analysts understand business requirements and translate them into application features. They communicate with users, document workflows, and help developers design appropriate solutions. For example, they may identify the information needed to register vehicles and determine how service requests should move from booking to delivery.

5. **Integration and Pro-Code Developer:**

Low-code applications sometimes require custom programming or external integrations. Developers with skills in JavaScript, Node.js, Python, APIs, and databases can extend the capabilities of low-code platforms. For example, a vehicle service application could connect to an external inventory system or use a custom API to calculate service estimates.

6. **Low-Code Consultant and Administrator:**

Consultants help organizations select and implement low-code platforms. Administrators manage users, permissions, application configuration, monitoring, and maintenance. They also help maintain data security and ensure that applications continue to operate reliably.

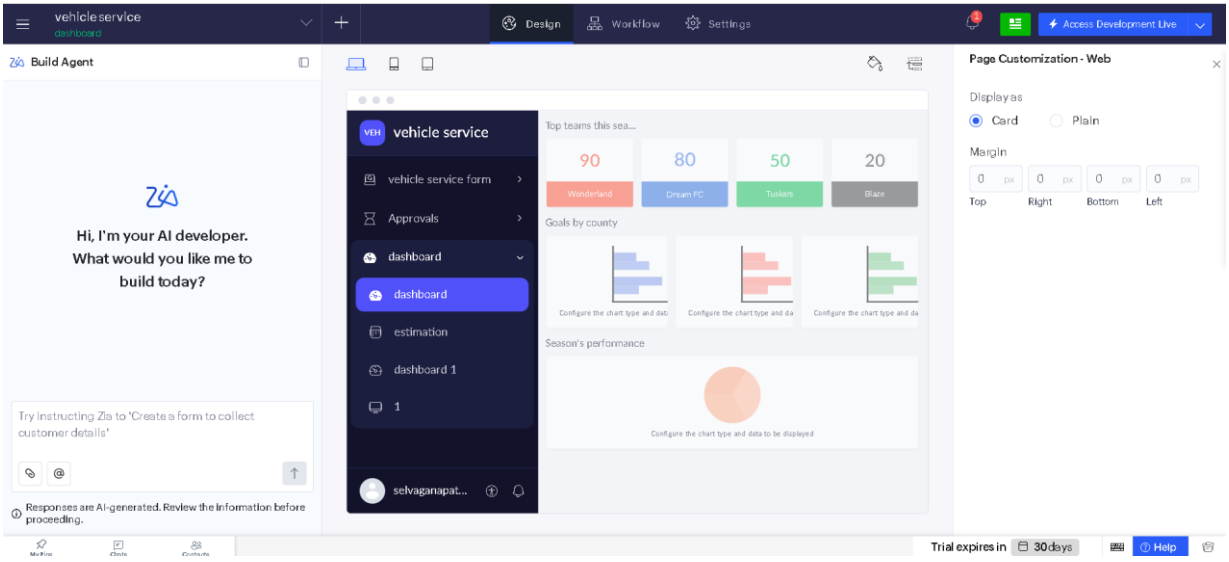
Skills Required for a Career in Low-Code Development

- Platform knowledge: Understanding tools such as Zoho Creator and their development features.
- Database fundamentals: Learning how to store, retrieve, and manage application data.
- Basic programming: Understanding logic, conditions, loops, and scripting languages such as Deluge.
- Problem-solving: Identifying business problems and creating suitable application solutions.
- Testing and debugging: Finding errors and ensuring that applications work as expected.

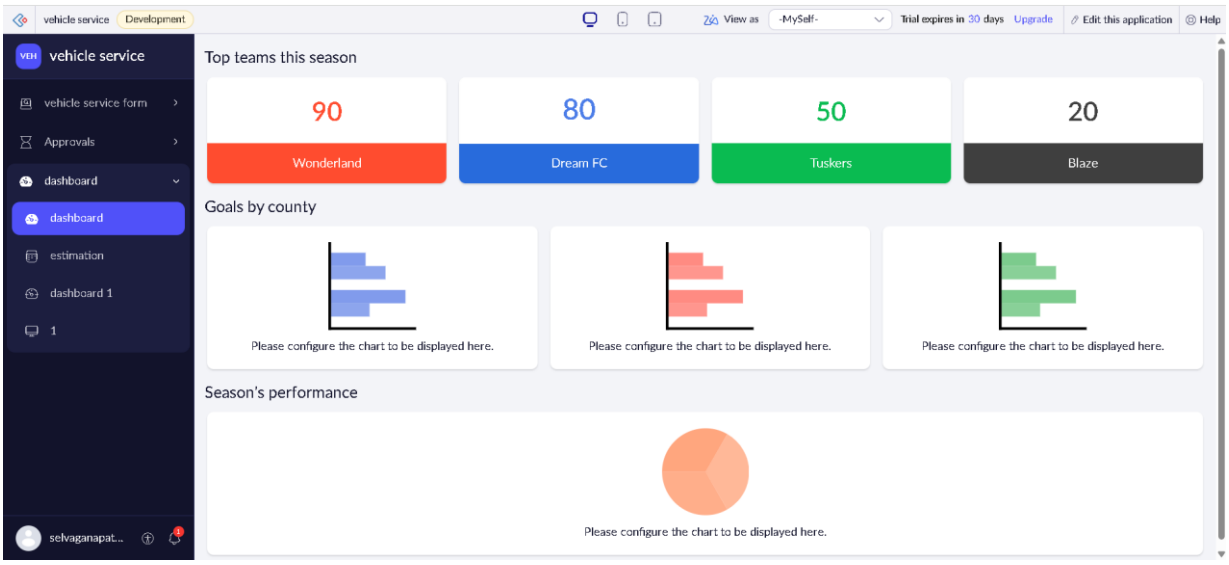
- Communication skills: Understanding user requirements and explaining technical solutions.

Future of Low-Code Development

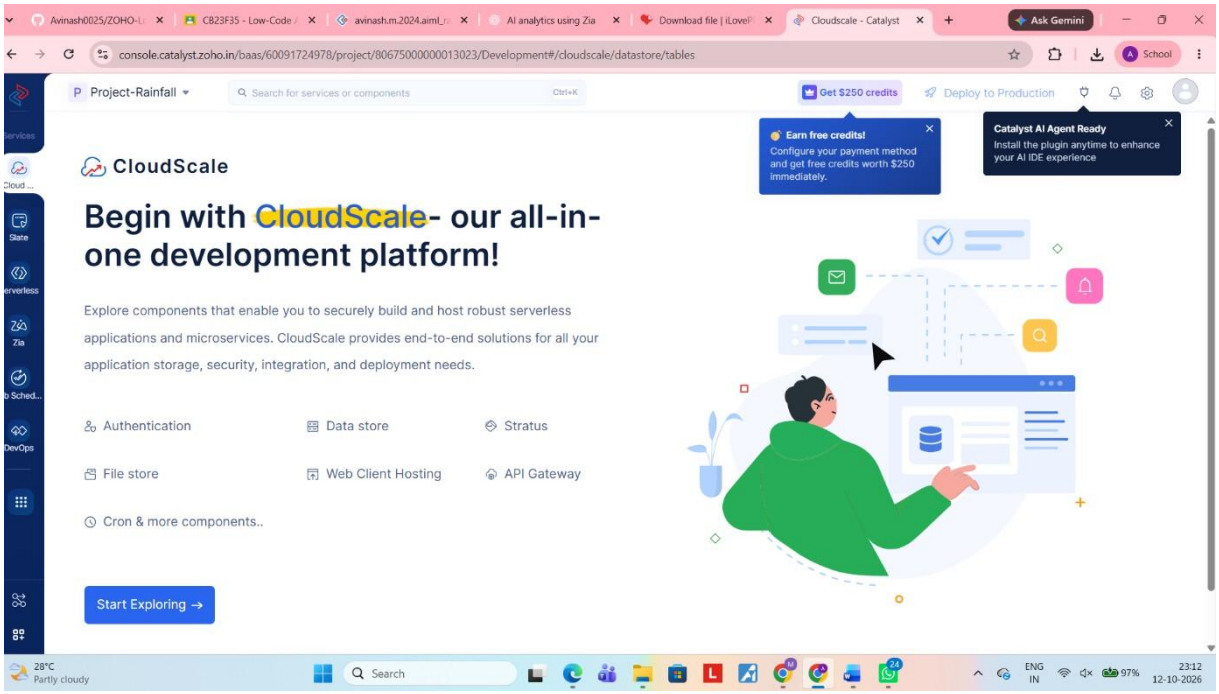
Integration with Artificial Intelligence: Low-code platforms are increasingly incorporating AI-assisted development, data analysis, and intelligent automation. These capabilities can help developers generate application components, analyze business data, and reduce manual work.



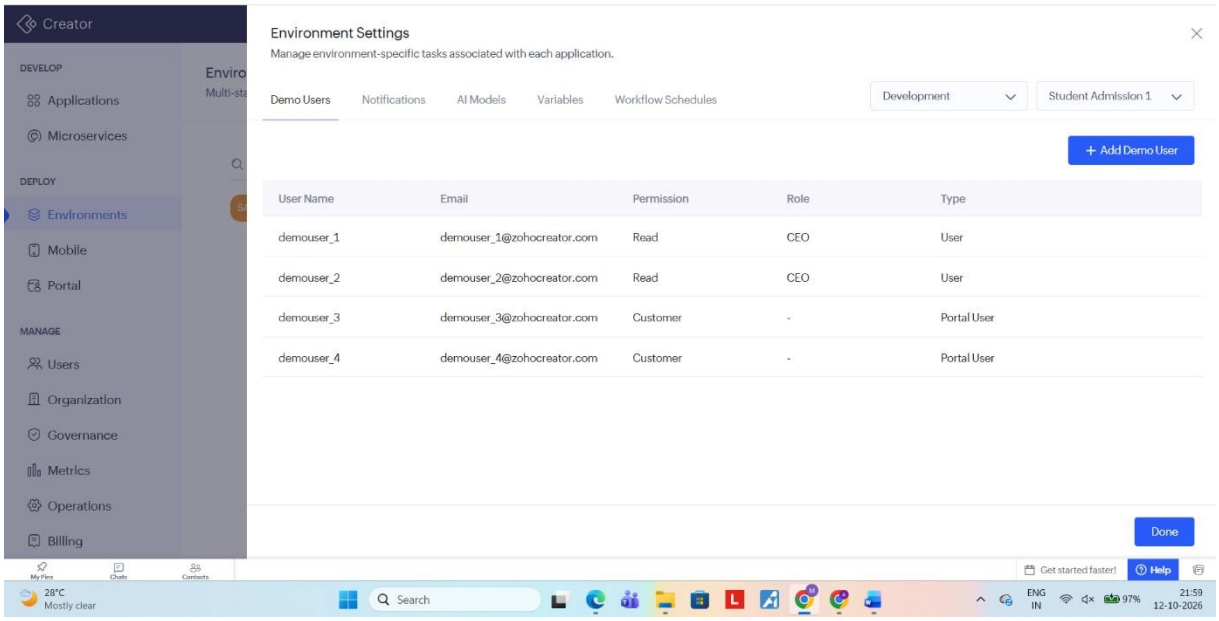
2. Growth of Business Automation: Organizations can use low-code tools to automate approvals, notifications, customer management, inventory processes, and reporting. This is expected to remain an important application area.



3. Combination of Low-Code and Pro-Code: Future applications will often combine visual development with custom programming. Low code platforms can handle standard forms and workflows, while traditional code can support specialized calculations, advanced integrations, and complex processing.

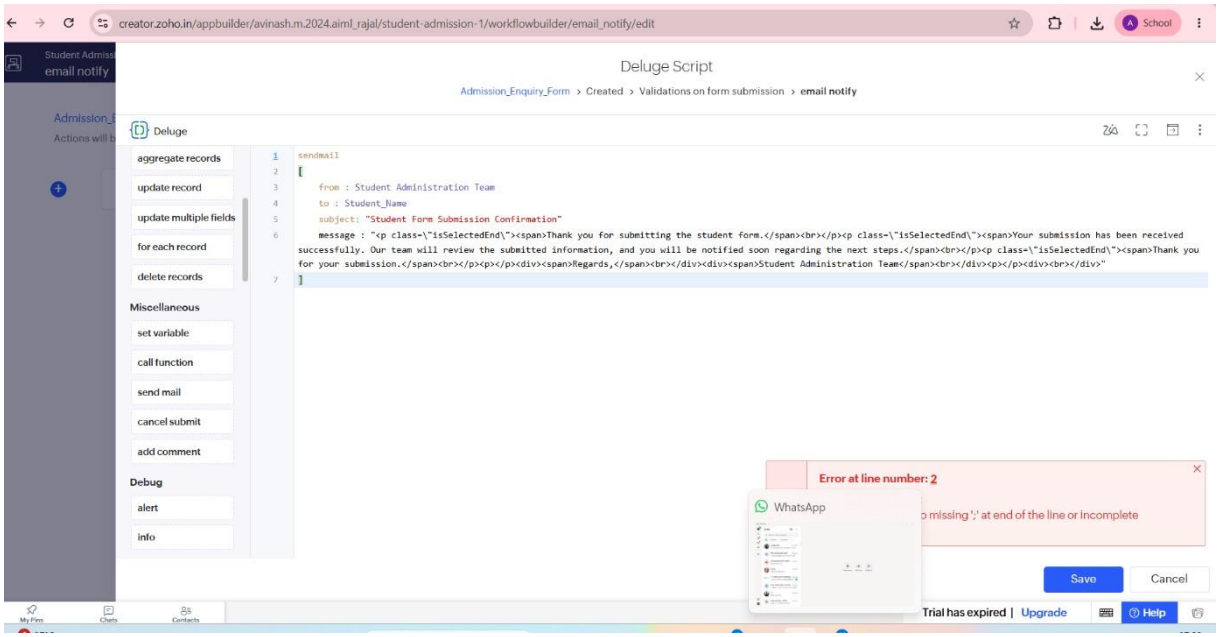


4. Greater Focus on Security and Governance: As organizations develop more applications, managing user permissions, protecting sensitive data, monitoring changes, and maintaining compliance will become increasingly important.



5. Cloud-Based Application Development: Cloud platforms make applications accessible across locations and devices. Low-code development can help organizations deploy and maintain business applications without managing every infrastructure component themselves.

6. Continued Need for Technical Skills: Low-code does not eliminate the need for programming knowledge. Developers must still understand application logic, data structures, testing, integration, and security to build reliable systems.



Conclusion:

Low-code development offers career opportunities in application development, workflow automation, software testing, consulting, and freelancing. These careers allow individuals to develop useful business applications with less manual coding and reduced development time.
Low-code platforms such as Zoho Creator are suitable for beginners because they provide visual development tools and built-in features for forms, reports, workflows, and dashboards. By gaining practical experience, learning database concepts, developing basic scripting skills, and completing real-world projects, students can build a strong portfolio and prepare for entry-level IT jobs. Furthermore, combining low-code knowledge with programming, API integration, and AI tools can help individuals expand their career opportunities and develop more advanced applications in the future.